

Flyhomes BBYS Calculator

Calculation Logic — Engineering Handoff Spec

Source of truth: `index.html` (single-file prototype), as implemented on 2026-08-11 · Verified against live JS, not just design docs

Purpose. This document specifies the calculation and decision logic exactly as implemented in the working prototype (`index.html`), so it can be re-implemented in production code. Where the prototype's behavior differs from — or isn't covered by — the design doc (`CLAUDE.md`), that is called out explicitly with a GAP or BUG tag. Everything else is marked MATCHES DOC.

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1. State model

A single mutable state object `S` drives every calculation. Relevant fields ([index.html:1148–1177](#)):

Field	Meaning	Default
<code>homeValue</code>	Departing home value (current home)	\$1,200,000
<code>mortgage</code>	Current mortgage balance on departing home	\$300,000
<code>newPrice</code>	Estimated purchase price of new home	\$800,000
<code>assets</code>	Liquid cash assets available	\$40,000
<code>transitionDays</code>	Days between buying new home and selling old one. User-editable, min 1	60
<code>gbc</code> , <code>cash</code> , <code>noCash</code> , <code>downsize</code>	Resolved product flags for the current render (derived — see §2)	—
<code>cashIntent</code> , <code>gbc_base</code> , <code>noCash_base</code>	Raw user intent flags, before Rule 3 auto-gating resolves them	—
<code>forceCashOffer</code>	"Option A" manual override: user chose standalone Cash Offer over an auto-recommended Cross Collateral. Reset to <code>false</code> whenever challenge selection changes	false
<code>naturalIsCC</code>	True if Cross Collateral would be recommended <i>without</i> the override — drives the switch-offer banner	—
<code>statesDiffer</code>	True when "current home state" and "buying state" dropdowns are both filled and unequal	false
<code>flyhomesLoan</code>	True = GBC bundle rate; false = GBC standalone rate	true
<code>savCashPct</code> / <code>savStagedPct</code> / <code>savHpaPct</code>	Savings-section sliders	3.5% / 5% / 4%
<code>savStagingCost</code>	Flat staging cost deducted from staged-home premium	\$3,500
<code>savMovingFee</code> , <code>savHousing</code> , <code>savStorage</code>	"Move Once" line items — auto-computed, user-editable	derived from <code>homeValue</code>

2. Challenge selection → product rule engine

Four challenge checkboxes map to internal ids: `dti` , `equity` , `cash` , `downsize` ([index.html:1640–1646](#)), stored in a `Set` called `selected` . Two functions resolve them into product flags on every change:

`syncStateFromChallenges()` then `applyRule3Gating()`. `applyRule3Gating()` also re-runs on every `render()` so slider changes (price, mortgage, home value) can flip the recommendation live.

2.1 `syncStateFromChallenges()` [index.html:1648–1662](#)

```
isDownsize      = selected.has('downsize')
isCrossCollateral = (selected.has('equity') && selected.has('cash')) || isDownsize

S.downsize      = isDownsize
S.noCash_base   = isCrossCollateral
S.cashIntent    = selected.has('cash') && !isCrossCollateral
S.gbc_base      = selected.has('dti') && !isCrossCollateral
S.forceCashOffer = false    // reset override on any challenge-card change
```

Matches design doc Rule 0 (downsize overrides) and Rule 1 (equity+cash → Cross Collateral).

MATCHES DOC

2.2 `applyRule3Gating()` [index.html:1667–1693](#)

```
isRule3CC = false
if (cashIntent && newPrice > 0):
    projectedFinalLTV = (newPrice + mortgage - homeValue * 0.9) / newPrice
    isRule3CC = projectedFinalLTV <= 0.75

if (statesDiffer):                                // Rule 5 – cross-state block
    noCash_base = false
    isRule3CC   = false
    if (!cashIntent): cashIntent = selected.has('cash')

naturalIsCC = noCash_base || isRule3CC

if (naturalIsCC && !downsize && forceCashOffer):
    // Option A: user overrode CC → standalone Cash Offer
    noCash = false; cash = true; gbc = gbc_base
else:
    noCash = naturalIsCC
    cash   = cashIntent && !isRule3CC
    gbc    = isRule3CC ? false : gbc_base    // CC can't pair with GBC
```

Matches design doc Rule 3 (auto-gate on `projectedFinalLTV`, with the "Keep More Cash" override) and Rule 5 (cross-state block, with fallback to Cash Offer + GBC / GBC-alone). **MATCHES DOC** — **except** for the downsize interaction described in §6.1 below.

2.3 Card visibility & labels — `applyProductLabel()` [index.html:1695–1748](#)

Condition	Result
<code>dtiOnly = selected.size===1 && has('dti')</code>	Hides main solution card, cost/benefit card, CTA card, number card. Shows only the GBC/DTI Buster card.
<code>noneSelected = selected.size===0</code>	Same as above — nothing shown.
<code>showGBC = !noneSelected && !isCrossCollat</code>	GBC (DTI Buster) card shows whenever <i>anything</i> is selected and the resolved product isn't Cross Collateral — regardless of whether "dti" was checked . This is what implements Rule 2/Rule 3's "+ DTI Buster" pairing.
<code>mostLikely = showGBC && !has('dti')</code>	"Most likely needed" badge on the DTI Buster card — shown when GBC is auto-added by Rule 2/3 rather than manually checked.
Label priority: <code>isDownsize</code> → <code>isCrossCollat</code> → <code>has('cash')</code> → <code>has('equity')</code>	First match wins for <code>solutionLabel</code> / <code>productLabel</code> / <code>intro</code> text. Downsize is checked first and unconditionally — see §6.1.

3. Core product formulas

Computed in `calc()` [index.html:1196–1264](#), branching on the resolved `S.noCash` / `S.cash` flags from §2:

```
availableEquity  = max(0, homeValue * 0.75 * 0.9 - mortgage)      // = Instant Equity value
combinedResources = assets + availableEquity
cashThreshold    = newPrice * 0.05
cashEligible     = combinedResources >= cashThreshold
gbcActive        = S.gbc && !noCash                               // GBC never pairs with Cross Collateral
```

3.1 Branch A — Cross Collateral (if `noCash` is true)

```
ccCltv = (newPrice + homeValue) * 0.75 - mortgage
maxLoan = max(0, min(newPrice * 1.05, ccCltv))
ltvAcq  = newPrice > 0 ? maxLoan / newPrice : 0
originationRate = ltvAcq > 0.90 ? 0.015 : 0.010
brokerFee      = maxLoan * 0.01

// Label:
if (S.downsize): solutionName = "Retire & Downsize"      productName = "Cross Collateral"
else:            solutionName = "Move with $0 out of pocket" productName = "Cross Collateral"
```

Matches design doc §5E Cross Collateral formula and §5B origination-fee tier.

MATCHES DOC

3.2 Branch B — Cash Offer (else if `cash && cashEligible`)

```
maxPurchasePrice = combinedResources / 0.05
maxLoan = max(0, min(newPrice, maxPurchasePrice) * 0.95)
ltv      = newPrice > 0 ? maxLoan / newPrice : 0
originationRate = ltv > 0.90 ? 0.015 : 0.010
brokerFee      = maxLoan * 0.01
solutionName = "All-Cash Advantage"  productName = "Cash Offer"
```

Matches design doc §5E "BBYS + Cash Offer" and Max Purchase Price formulas.

MATCHES DOC

3.3 Branch C — Instant Equity (else — default / fallback)

```
maxLoan = max(0, homeValue * 0.75 * 0.9 - mortgage)
originationRate = 0.02          // 1st lien only – see §6.2
brokerFee      = maxLoan * 0.005
solutionName = "Equity for Down Payment"  productName = "Instant Equity"
```

Matches design doc §5E Instant Equity formula and §5B (1st-lien rate).
not implemented; see §6.2.

MATCHES DOC

— 2nd-lien rate is documented but

This branch is also the fallback whenever `cash` is true but `cashEligible` is false (insufficient combined cash + equity for the 5% down payment) — in that case a "cash warning" banner is additionally shown ([index.html:1598–1604](#)) with the shortfall amount: `gap = cashThreshold - combinedResources`.

4. Fee formulas

4.1 Origination fee

```
originationFee = maxLoan * originationRate    // rate resolved per branch above, §3
```

4.2 Broker fee

Rolled into "Other Costs" for display — never shown as a standalone line item to the buyer, per design doc §5D. Rate depends on product track (see §3): Cross Collateral 1% of capped loan, Cash Offer 1% of FCO loan amount, Instant Equity 0.5% of advance amount.

4.3 GBC fee — `getGbcFee(homeValue, isBundle)` [index.html:1182–1193](#)

```
if      (homeValue <= 500,000):  standalone = 2,500;  credit = 0
elif    (homeValue <= 750,000):  standalone = 3,500;  credit = 1,000
elif    (homeValue <= 1,000,000): standalone = 5,000;  credit = 2,500
elif    (homeValue <= 1,500,000): standalone = 7,500;  credit = 2,500
elif    (homeValue <= 2,000,000): standalone = 10,000; credit = 2,500
else:                                     return null    // >$2M exception — manual review required

return isBundle ? (standalone - credit) : standalone
```

`isBundle` passed in as `!dtiOnly && S.flyhomesLoan` — i.e. the bundle (net) rate only applies when a Flyhomes loan product is shown alongside GBC *and* it isn't the DTI-only scenario. In DTI-only mode, GBC is always priced standalone (no loan bundle exists to net against). **MATCHES DOC** §5C exactly, including the >\$2M exception returning `null` and being surfaced in the UI as "Exception — contact us" rather than a computed number.

4.4 Accrued interest

```
accruedInterest = maxLoan * 0.0999 * (transitionDays / 365)
```

Only shown/rendered when `maxLoan > 0` (hidden for DTI-only). **MATCHES DOC**

4.5 Total estimated cost

```
gbcFeeNum = (gbcFee === null) ? 0 : (gbcFee ?? 0)    // >$2M exception treated as $0 in the t
total      = originationFee + brokerFee + gbcFeeNum + accruedInterest
```

Note: when the GBC fee is an unresolved >\$2M exception, the displayed total silently excludes it (treated as \$0) even though the UI separately shows "Exception — contact us" for the GBC line. Production implementation should decide whether the grand total should also flag as "incomplete/exception" rather than silently under-totaling.

5. Estimated Savings formulas

Computed in `calcSavings()` [index.html:1325–1337](#):

5.1 Cash Offer Price Advantage

```
isCash      = (cash && assets >= newPrice * 0.05) || noCash
cashDiscount = isCash ? newPrice * (savCashPct / 100) : 0
```

Gap vs. design doc: **GAP** the design doc (§5F.1) states this row shows whenever `isCashOfferSelected`, or the recommended product is Cross Collateral, or `forceCashOffer` is true. The actual eligibility check additionally requires `assets >= newPrice * 0.05` — i.e. *liquid cash alone* must cover the 5% down payment, not the combined cash+equity figure (`combinedResources`) used everywhere else in the calculator for cash eligibility. A user who is "cash eligible" for the main Cash Offer product (via equity) can still have the savings-section cash-discount row hidden. Not documented in CLAUDE.md — confirm intended behavior with product before porting.

5.2 Staged Home Premium

```
staged = max(0, homeValue * (savStagedPct / 100) - savStagingCost) // savStagingCost default
```

Matches design doc tooltip (net of \$3,500 staging cost) — note the doc's "Formula" line omits the staging-cost subtraction and only mentions it in the tooltip text; the code implements the tooltip version. **MATCHES DOC** (tooltip version)

5.3 Home Price Appreciation (HPA)

```
hpa = newPrice * (savHpaPct / 100) * (transitionDays / 30) / 12
```

Algebraically identical to design doc's `newHomePrice * (hpaRate% / 12) * (transitionDays / 30)` — order of operations differs, result is the same. Buy-side only, per doc. **MATCHES DOC**

5.4 "Move Once, Not Twice" (logistics)

```
// Auto-computed defaults (index.html:1397–1400), independently user-editable after:
savMovingFee = round(homeValue * 0.003 + 500)
savHousing   = round(homeValue * 0.006 * (transitionDays / 30))
savStorage   = round((150 + homeValue * 0.0002) * (transitionDays / 30))
logistics    = savMovingFee + savHousing + savStorage // uses live (possibly edited) field v
```

MATCHES DOC exactly.

5.5 Grand total

```
total = cashDiscount + staged + hpa + logistics
```

The DTI-only ("GBC-only") card computes its own total that always excludes the cash discount row (since Cash Offer is never active in DTI-only mode): `g-total = staged + hpa + logistics`.

6. Known issues & discrepancies vs. design doc

6.1 **BUG** Downsize + cross-state produces mismatched label vs. math

Design doc Rule 0 states downsize "overrides all other logic combinations." In the actual implementation, Rule 5's cross-state block runs *after* Rule 0 and unconditionally zeroes `noCash_base` — including when `S.downsize` is true:

```
// applyRule3Gating(), index.html:1673–1678
if (S.statesDiffer) {
  S.noCash_base = false; // <- forced false even if downsize selected
  isRule3CC      = false;
  ...
}
```

Result: if a user selects "Want to downsize mortgage-free" (Option 4) *and* the current-home state differs from the buying state, `S.noCash` ends up `false` and `calc()` silently falls through to the **Instant Equity** branch (§3.3) for all numbers (max loan, fees, total cost). But `applyProductLabel()` checks `S.downsize` first and unconditionally ([index.html:1733](#)), so the UI still displays the **"Retire & Downsize" / "Cross Collateral"** title, subtitle, and the "same state required" note — while every dollar figure on the card is actually computed as Instant Equity.

Recommendation: decide product intent before porting — either (a) Cross Collateral truly requires same-state homes and downsize should fall back to a different product/labeling when states differ, or (b) downsize should be exempted from the cross-state block per Rule 0's stated priority. Currently neither is true; the label and the math disagree.

6.2 **GAP** Instant Equity 2nd-lien rate not implemented

Design doc §5B specifies 2.0% for 1st lien and 2.5% for 2nd lien on the Instant Equity track. The code hardcodes `originationRate = 0.02` with an inline comment "1st lien; 2nd lien (2.5%) requires LO confirmation" ([index.html:1245](#)) — there is no lien-position input or branch anywhere in the app. Production build needs to decide how lien position is captured (if at all) before this rate can vary.

6.3 GAP Cash-discount eligibility differs between main card and savings section

See §5.1 above — the savings section's cash-discount visibility uses raw `assets` , not `combinedResources` (cash + unlockable equity) used for the main product's `cashEligible` . Not documented anywhere in CLAUDE.md.

6.4 GAP Dead code: unused `savings` field in `calc()`

```
savings = max(0, maxLoan * 0.04 - total)    // index.html:1260
```

This value is returned from `calc()` but never read or rendered anywhere in the app. Likely a leftover from an earlier design iteration (an implied "4% return" comparison). Safe to drop when porting unless product intends to revive it — flag to product/design before deciding.

6.5 GAP GBC fee >\$2M exception silently excluded from grand total

See §4.5 — the UI shows "Exception — contact us" for the GBC line item itself, but the overall Total Estimated Cost treats the unresolved exception as \$0 rather than flagging the total as incomplete.

7. Appendix: constants reference

Constant	Value	Used in
Accrued interest rate	9.99% / yr	Total cost — all tracks
Instant Equity multiplier	$0.75 \times 0.9 = 0.675$	Instant Equity value / available equity
Cash Offer max LTV	95%	Cash Offer max loan cap
Cash Offer min down payment	5%	Cash eligibility threshold
Cross Collateral acquiring LTV cap	105%	Cross Collateral max loan cap
Cross Collateral combined LTV	75%	Cross Collateral CLTV formula
Origination fee — CO / CC, LTV > 90%	1.5%	Origination fee
Origination fee — CO / CC, LTV ≤ 90%	1.0%	Origination fee
Origination fee — Instant Equity (1st lien)	2.0%	Origination fee (2nd lien 2.5% not implemented — \$6.2)
Broker fee — Cross Collateral	1% of capped loan	Rolled into Other Costs
Broker fee — Cash Offer	1% of FCO loan amount	Rolled into Other Costs
Broker fee — Instant Equity	0.5% of advance amount	Rolled into Other Costs
Rule 3 gating threshold	$\text{projectedFinalLTV} \leq 0.75 \rightarrow \text{Cross Collateral}$	Auto-recommendation between CC and CO
Staging cost (flat, editable)	\$3,500	Staged Home Premium
Default sliders	Cash discount 3.5%, Staged premium 5%, HPA 4%	Estimated Savings
Default transition days	60 (min 1)	Accrued interest, HPA, temp housing, storage

Generated from `index.html` for engineering handoff. Line-number references are point-in-time as of 2026-08-11 and will drift as the file changes — re-verify against source before treating as final spec.